Information Visualization 01
What-Why

Patrick Riehmann

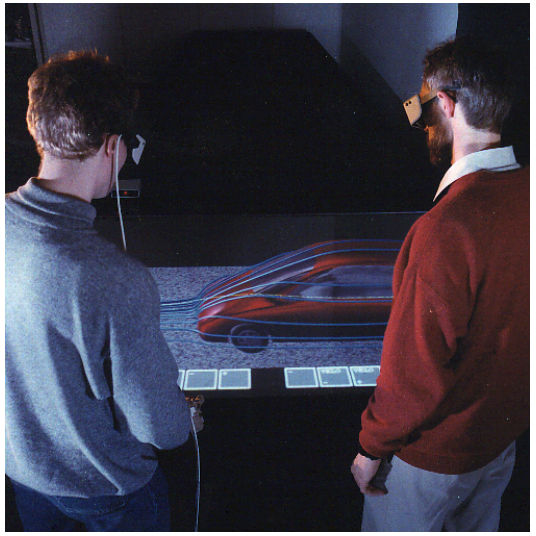
The ability to take data—to be able to **understand** it, to **process** it, to **extract value** from it, to **visualize** it, to **communicate** it—that's going to be a hugely important skill in the next decades, … because now we really do have **essentially free and ubiquitous data**. So the complimentary scarce factor is the ability to understand that data and extract value from it.

Hal Varian, Google's Chief Economist

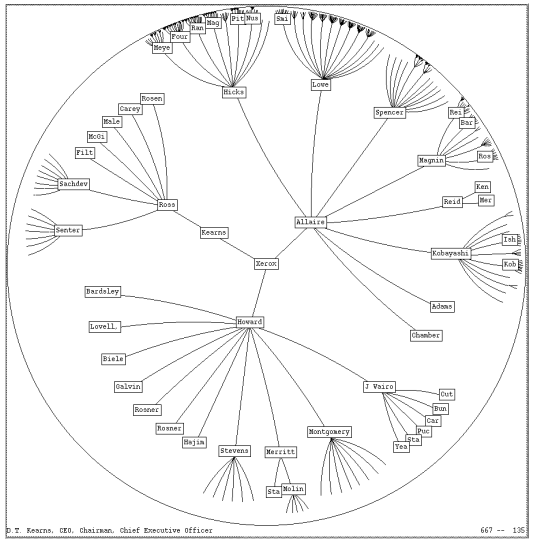
The McKinsey Quarterly, Jan 2009

Visualization: Terminology

- Scientific Visualization
  - Physical / simulated data
  - Data
    - Automotive
    - Weather
    - Medical
    - ...

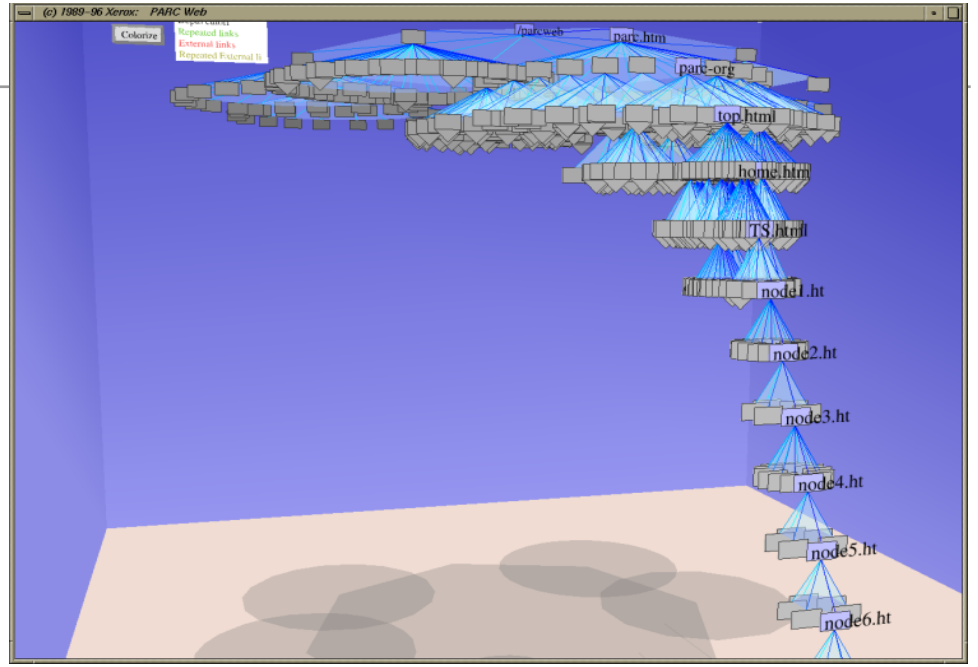


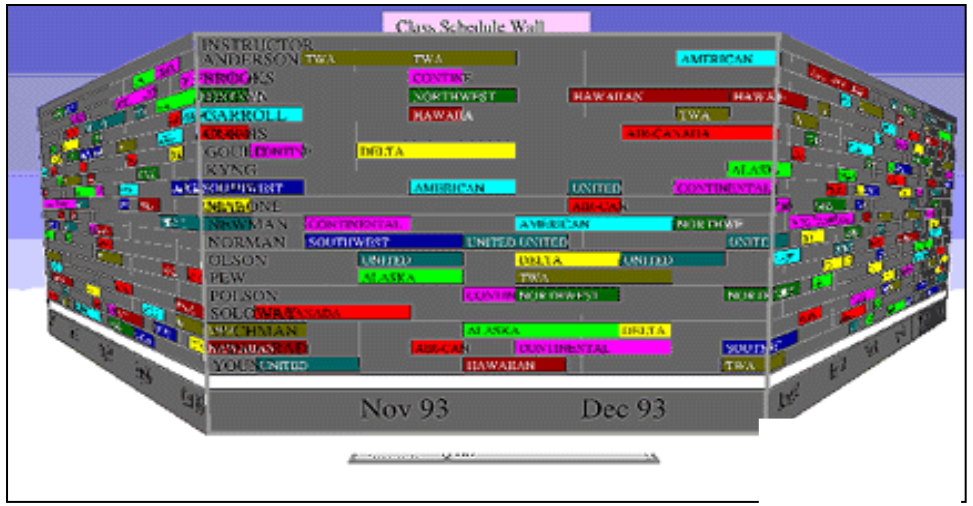
- Information Visualization
  - Abstract, non-physical data
  - Data
    - Financial/business
    - Statistics
    - Software
    - Text
    - W W W
    - ...



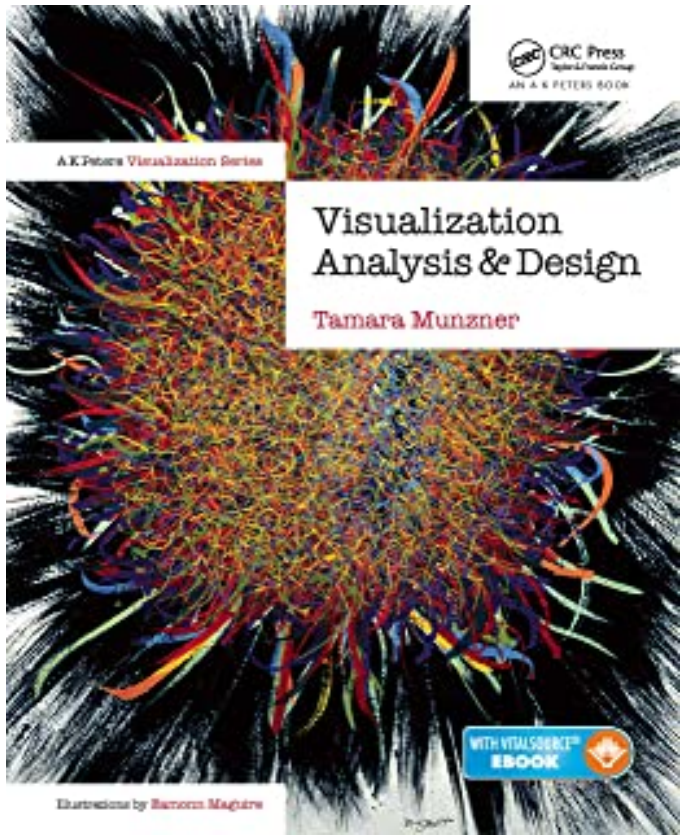
Information Visualization

- In 1989 Xerox PARC Jock Mackinlay, Stu Card, and George Robertson coined the term “Information Visualization”
- What is “Information”?
  - Items, entities, things without a direct physical correspondence
  - Notion of abstractness of the entities is important
- What is “visualization”?
  - The use of computer-supported, interactive visual representations of data to amplify cognition. [Card, Mackinlay Shneiderman ‘98]
- Visuals help us think, they support discovery, decision making, explanation, …





- First three lectures are based on slides and talks by T. Munzner and her book: Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014.



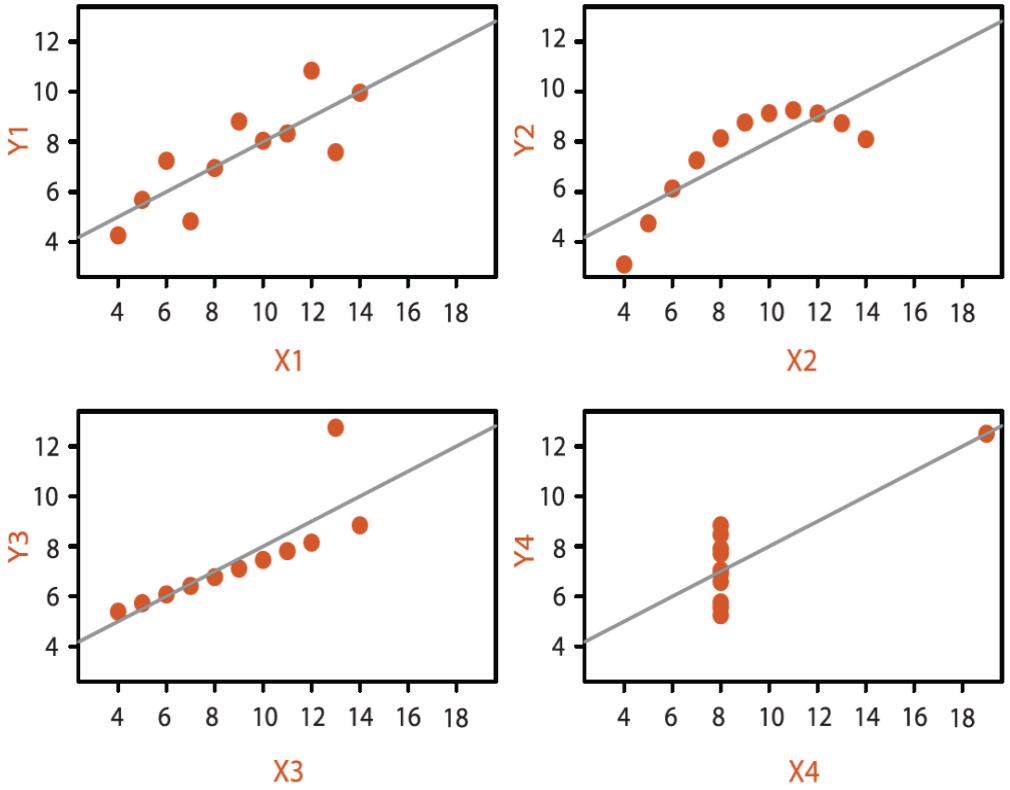
Munzner's Definition of Visualization

- Computer-based visualization systems provide visual representations of datasets designed to help people carry out tasks more effectively.
- Visualization is suitable when there is a need to augment human capabilities rather than replace people with computational decision-making methods.
- Don't need visualization if a fully automatic solution exists and is trusted
- Need human in the loop since many analysis problems **ill-specified**
  - Don't know exactly what to ask in advance
  - Need to browse, explore, look for patterns, outliers …
  - Speed up and improve upon analysis and decision-making processes!
- Important uses of visualization
  - Regular use by end users e.g. for exploratory analysis of scientific data, live monitoring, …
  - Presentation of known results – data stories, data journalism etc.
  - Verify statistical analysis and derived models
  - …

Why Show All the Data?

- Summaries lose information, the details matter
  - Need to confirm expected and find unexpected patterns
  - Assess validity of statistical model

Anscombe's Quartet

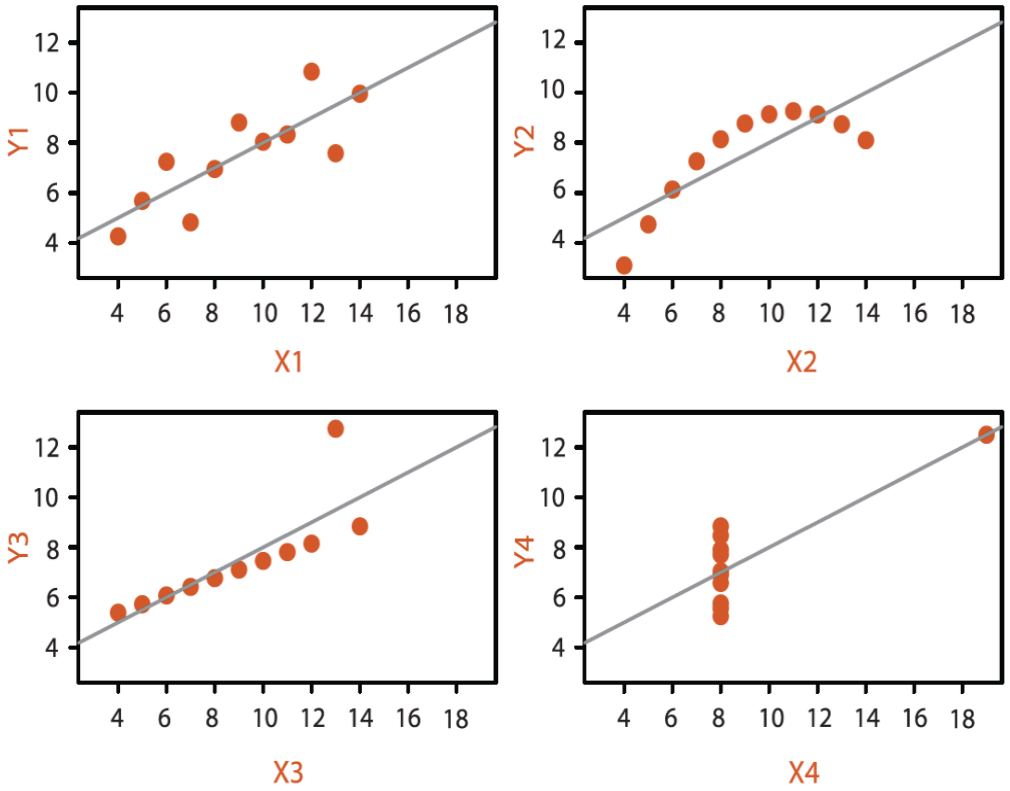


Identical statistics

x mean	9
x variance	10
y mean	8
y variance	4
x/y correlation	0.816

Anscombe‘s Advice

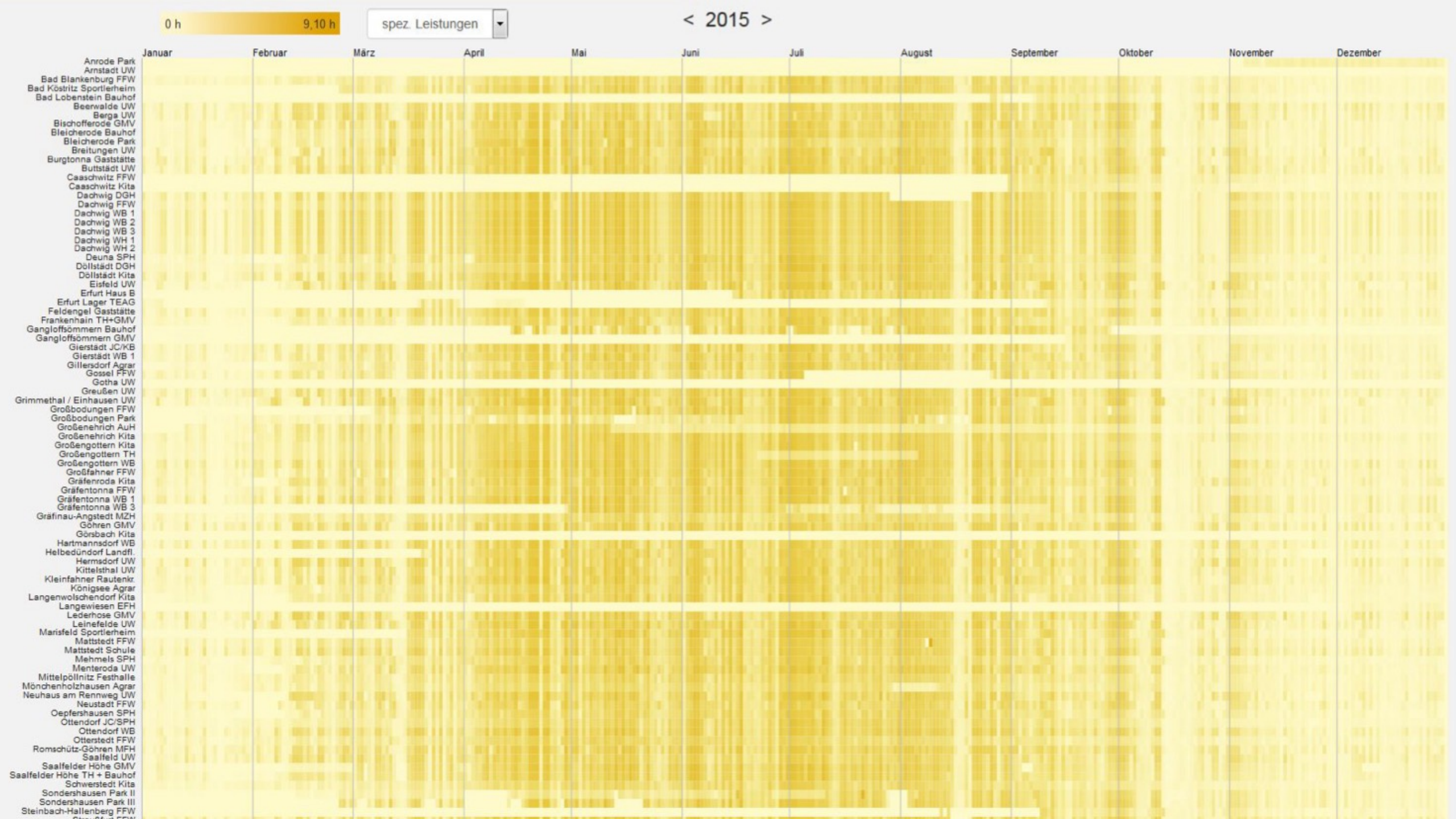
“...make both calculations and graphs. Both sorts of output should be studied; each will contribute to understanding.”



Identical statistics	
x mean	9
x variance	10
y mean	8
y variance	4
x/y correlation	0.816

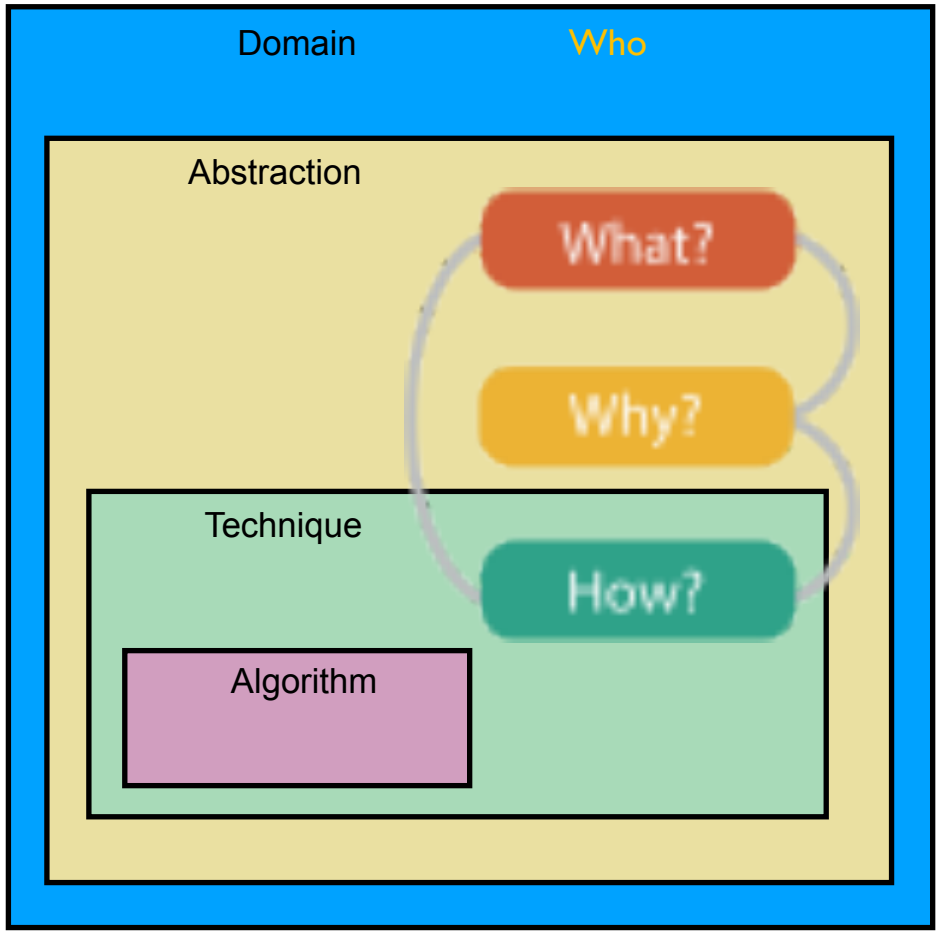
Why Use Visual Representations?

- Human visual system is high-bandwidth channel to brain
  - Overview possible due to background processing
    - Subjective experience of seeing everything simultaneously
    - Significant processing occurs in parallel and pre-attentively
  - Visual representation can serve as effective externalization of mental models – replaces partially cognition of numbers, relationships, … by perception
- Sound: lower bandwidth and different semantics
  - Subjective experience of sequential stream (overview not supported)
- Touch/haptics: impoverished record/replay capacity
  - Only low-bandwidth communication thus far
- Taste, smell: no viable record/replay devices


0 h
9,10 h
spez. Leistungen
< 2015 >
Januar
Februar
März
April
Mai
Juni
Juli
August
September
Oktober
November
Dezember
Anrode Park
Arnstadt UW
Bad Blankenburg FFW
Bad Köstritz Sportlerheim
Bad Lobenstein Bauhof
Beerwalde UW
Berga UW
Bischofferode GMV
Bleicherode Bauhof
Bleicherode Park
Breitungen UW
Burgtonna Gaststätte
Buttstädt UW
Caaschwitz FFW
Caaschwitz Kita
Dachwig DGH
Dachwig FFW
Dachwig WB 1
Dachwig WB 2
Dachwig WB 3
Dachwig WH 1
Dachwig WH 2
Deuna SPH
Döllstädt DGH
Döllstädt Kita
Eisfeld UW
Erfurt Haus B
Erfurt Lager TEAG
Feldengel Gaststätte
Frankenhain TH+GMV
Gangloffsömmern Bauhof
Gangloffsömmern GMV
Gierstädt JC/KB
Gierstädt WB 1
Gillersdorf Agrar
Gossel FFW
Gotha UW
Greußen UW
Grimmethal / Einhausen UW
Großbodungen FFW
Großbodungen Park
Großenehrich AuH
Großenehrich Kita
Großengottern Kita
Großengottern TH
Großengottern WB
Großfahner FFW
Gräfenroda Kita
Gräfentonna FFW
Gräfentonna WB 1
Gräfentonna WB 3
Gräfinau-Angstedt MZH
Göhren GMV
Görsbach Kita
Hartmannsdorf WB
Helbedündorf Landfl.
Hermsdorf UW
Kittelsthal UW
Kleinfahner Rautenkr.
Königsee Agrar
Langenwolschendorf Kita
Langewiesen EFH
Lederhose GMV
Leinefelde UW
Marisfeld Sportlerheim
Mattstedt FFW
Mattstedt Schule
Mehmels SPH
Menteroda UW
Mittelpöllnitz Festhalle
Mönchenholzhausen Agrar
Neuhaus am Rennweg UW
Neustadt FFW
Oepfershausen SPH
Ottendorf JC/SPH
Ottendorf WB
Otterstedt FFW
Romschütz-Göhren MFH
Saalfeld UW
Saalfelder Höhe GMV
Saalfelder Höhe TH + Bauhof
Schwerstedt Kita
Sondershausen Park II
Sondershausen Park III
Steinbach-Hallenberg FFW

Analysis Framework: Four Levels, Four Questions

- *Domain* situation
  - **Who** are the target users?
- *Abstraction*
  - Translate from specifics of domain to vocabulary of visualization
  - **Data abstraction**: **What** is shown?
  - **Task abstraction: Why** is the user looking at it?
- *Technique (Munzner: Idiom)*
  - **How** is the data shown and explored?
    - **Visual encoding** technique: how to visually represent data
    - **Interaction** technique: how to manipulate/explore
- *Algorithm*
  - Efficient computation and rendering



[A Nested Model of Visualization Design and Validation. Munzner. IEEE TVCG 15(6):921-928, 2009 (Proc. InfoVis 2009).]

[A Multi-Level Typology of Abstract Visualization Tasks Brehmer and Munzner. IEEE TVCG 19(12):2376-2385, 2013 (Proc. InfoVis 2013).]

Analysis: What, Why and How

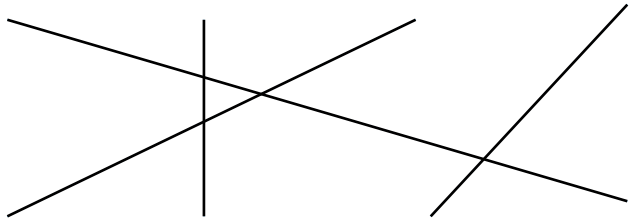
- **What** is shown?
  - **Data** abstraction
- **Why** is the user looking at it?
  - **Task** abstraction
- **How** is it shown?
  - **Technique**: visual encoding and interaction



- Abstract vocabulary avoids domain-specific terms
  - Translation process iterative and often complex
- It helps to think systematically about this huge design space and the choices
- Analyzing existing designs as an inspiration for designing new

Quiz

- How – What – Why – Who



- Task – Data – Users - Technique

What?

Why?

How?

What?

Datasets

Data Types

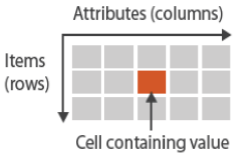
→ Items → Attributes → Links → Positions → Grids

Data and Dataset Types

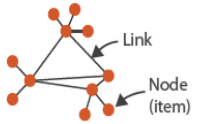
Tables	Networks & Trees	Fields	Geometry	Clusters, Sets, Lists
Items	Items (nodes)	Grids	Items	Items
Attributes	Links	Positions	Positions	
	Attributes	Attributes		

Dataset Types

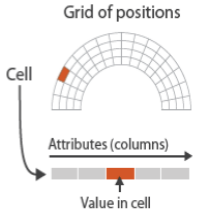
→ Tables



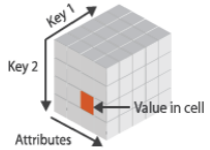
→ Networks



→ Fields (Continuous)



→ *Multidimensional Table*



→ *Trees*



→ Geometry (Spatial)



Dataset Availability

→ Static



→ Dynamic



Attributes

Attribute Types

→ Categorical



→ Ordered

→ *Ordinal*



→ *Quantitative*



Ordering Direction

→ Sequential



→ Diverging

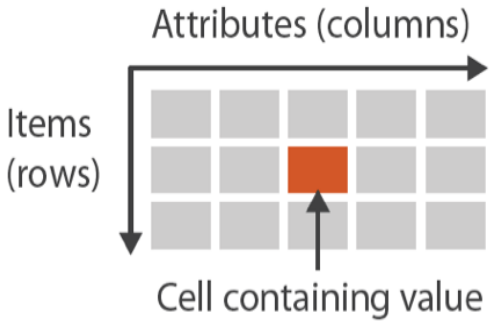


→ Cyclic

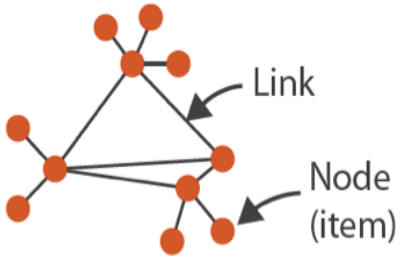


Dataset Types

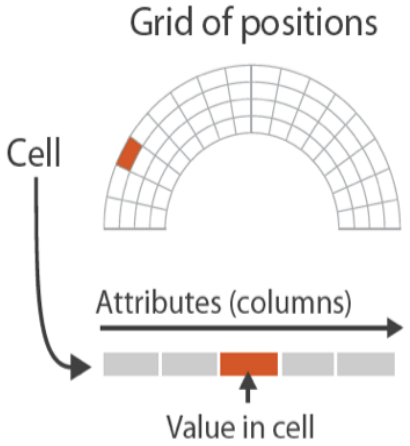
➔ Tables



➔ Networks

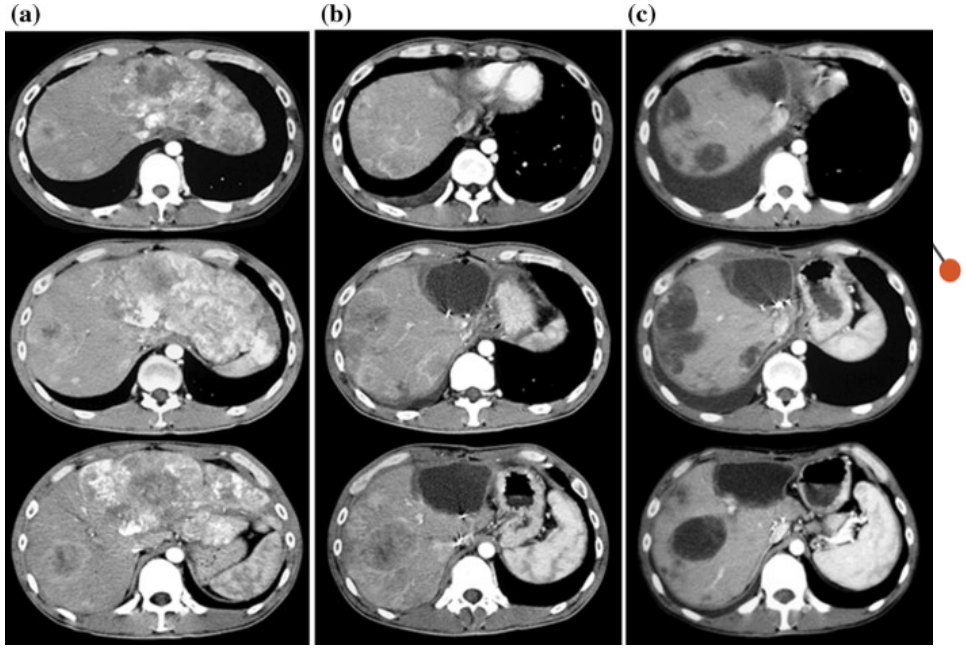


➔ Fields (Continuous)

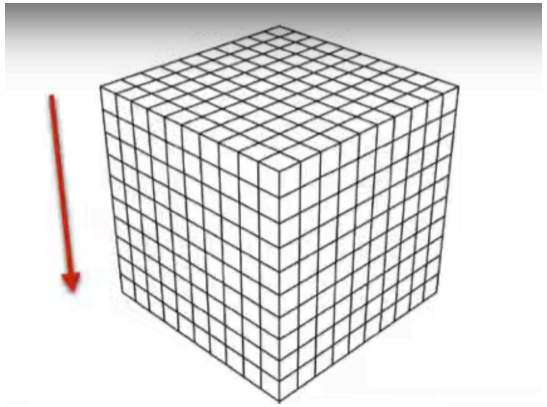


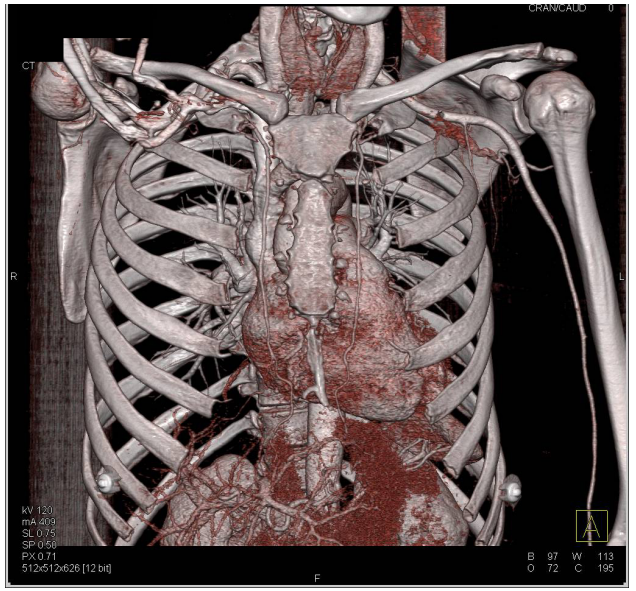
➔ Geometry (Spatial)





https://media.springernature.com/lw685/springer-static/image/art%3A10.1007%2Fs00268-019-05052-5/MediaObjects/268_2019_5052_Fig1_HTML.jpg



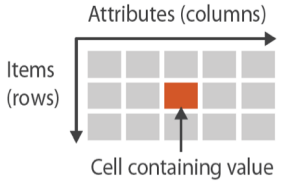


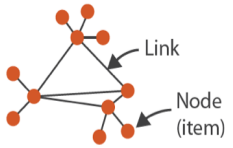
https://ctisus.com/resources/library/teaching-files/vascular/254181.jpg

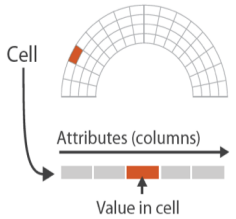
Datasets Described by Various Data Types

Data and Dataset Types

Tables	Networks & Trees	Fields	Geometry	Clusters, Sets, Lists
Items	Items (nodes)	Grids	Items	Items
Attributes	Links	Positions	Positions	
	Attributes	Attributes		









Dataset Availability

- Static



- Dynamic



Table

A	B	C	S	T	U
Order ID	Order Date	Order Priority	Product Container	Product Base Margin	Ship Date
3	10/14/06	5-Low	Large Box	0.8	10/21/06
6	2/21/08	4-Not Specified	Small Pack	0.55	2/22/08
32	7/16/07	2-High	Small Pack	0.79	7/17/07
32	7/16/07	2-High	Jumbo Box		7/17/07
32	7/16/07	2-High	Medium Box		7/18/07
32	7/16/07	2-High	Medium Box		7/18/07
35	10/23/07	4-Not Specified	Wrap Bag	0.52	10/24/07
35	10/23/07	4-Not Specified	Small Box	0.58	10/25/07
36	11/3/07	1-Urgent	Small Box	0.55	11/3/07
65	3/18/07	1-Urgent	Small Pack	0.49	3/19/07
66	1/20/05	5-Low	Wrap Bag	0.56	1/20/05
69	5	4-Not Specified	Small Pack	0.44	6/6/05
69	5	4-Not Specified	Wrap Bag	0.6	6/6/05
70	12/18/06	5-Low	Small Box	0.59	12/23/06
70	12/18/06	5-Low	Wrap Bag	0.82	12/23/06
96	4/17/05	2-High	Small Box	0.55	4/19/05
97	1/29/06	3-Medium	Small Box	0.38	1/30/06
129	11/19/08	5-Low	Small Box	0.37	11/28/08
130	5/8/08	2-High	Small Box	0.37	5/9/08
130	5/8/08	2-High	Medium Box	0.38	5/10/08
130	5/8/08	2-High	Small Box	0.6	5/11/08
132	6/11/06	3-Medium	Medium Box	0.6	6/12/06
132	6/11/06	3-Medium	Jumbo Box	0.69	6/14/06
134	5/1/08	4-Not Specified	Large Box	0.82	5/3/08
135	10/21/07	4-Not Specified	Small Pack	0.64	10/23/07
166	9/12/07	2-High	Small Box	0.55	9/14/07
193	8/8/06	1-Urgent	Medium Box	0.57	8/10/06
194	4/5/08	3-Medium	Wrap Bag	0.42	4/7/08

attribute

item

cell

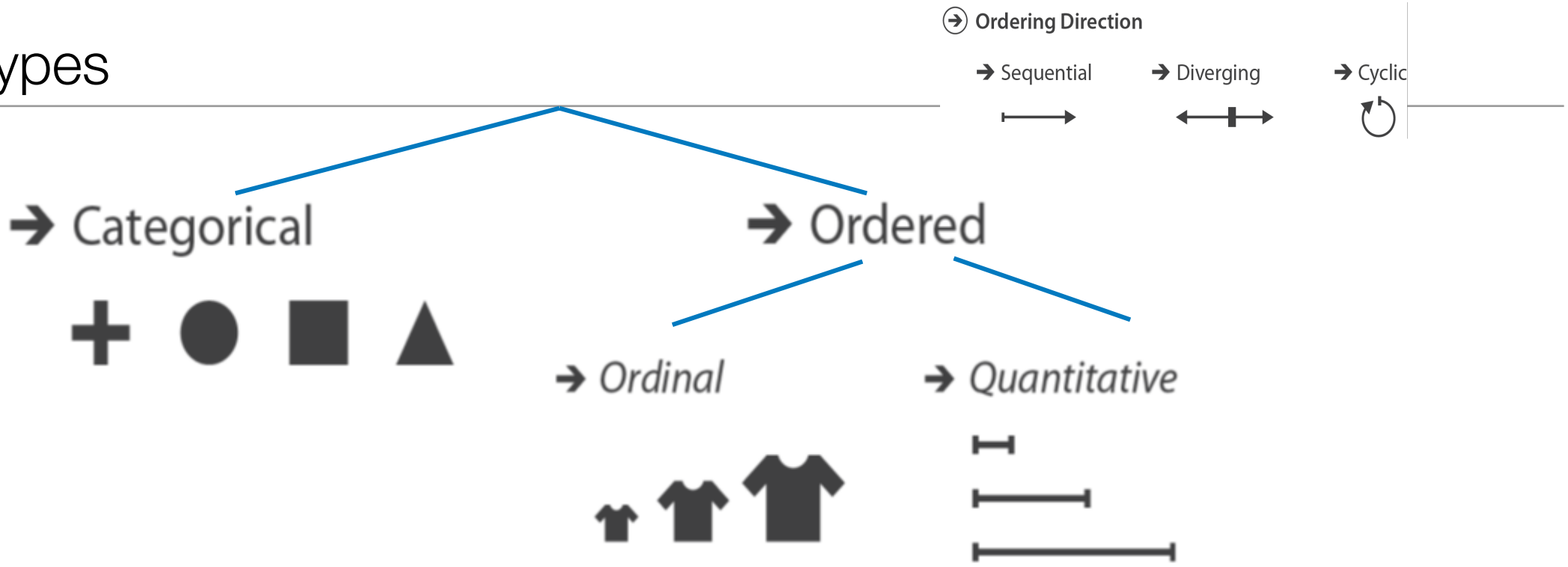
In each cell, an attribute value is stored

Data Types

➔ Items ➔ Attributes ➔ Links ➔ Positions ➔ Grids

- Item: individual discrete entity
  - A row in a simple table or a node in a network.
  - E.g. items may be people, stocks, coffee shops, genes, or cities.
- Attribute: specific property that can be measured, observed, or logged
  - Synonyms for attribute are variable and data dimension
  - ex: salary, price, number of sales, protein expression levels, or temperature.
- Link: specifies relationship between items, typically within a network.
- Position: spatial data, providing a location in two-dimensional (2D) or three-dimensional (3D) space – very special attribute.
  - Ex: a position might be a latitude–longitude pair describing a location on the Earth's surface or three numbers specifying a location within the region of space measured by a medical scanner.
- Grid: specifies the strategy for sampling continuous data in terms of both geometric and topological relationships between its cells.

Attribute Types



- Categorical data (nominal data)
  - No implicit ordering, often hierarchies of categories
  - Ex: fruit names, movie genres, file types and city names

- Ordered data has well-defined ordering

-Ordinal data has ordering
but no arithmetic possible

-Quantitative data allows computation

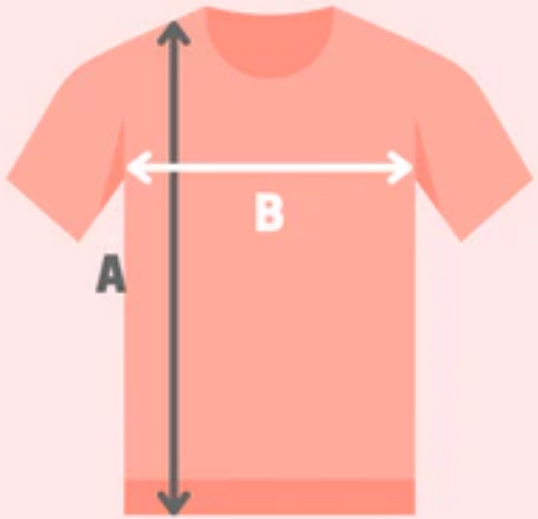
- Integer and real numbers: measurements (temp,..), number of something, …

Cardinality and Range

- Cardinality refers to the number of unique values that an attribute can take
  - Low Cardinality: When an attribute has only a few unique values. For example, the attribute "Gender" with values like "Male" and "Female" has low cardinality because there are very few unique values (2 in this case)
  - High Cardinality: When an attribute has many unique values. For example, an attribute like "Customer ID" or "Social Security Number" typically has high cardinality because there are as many unique values as there are entities (e.g., each customer has a unique ID)
- Vis may work better for attributes with low cardinality (e.g., bar charts) versus high cardinality (e.g., scatter plots)

Cardinality and Range

DAD-T-SHIRTS

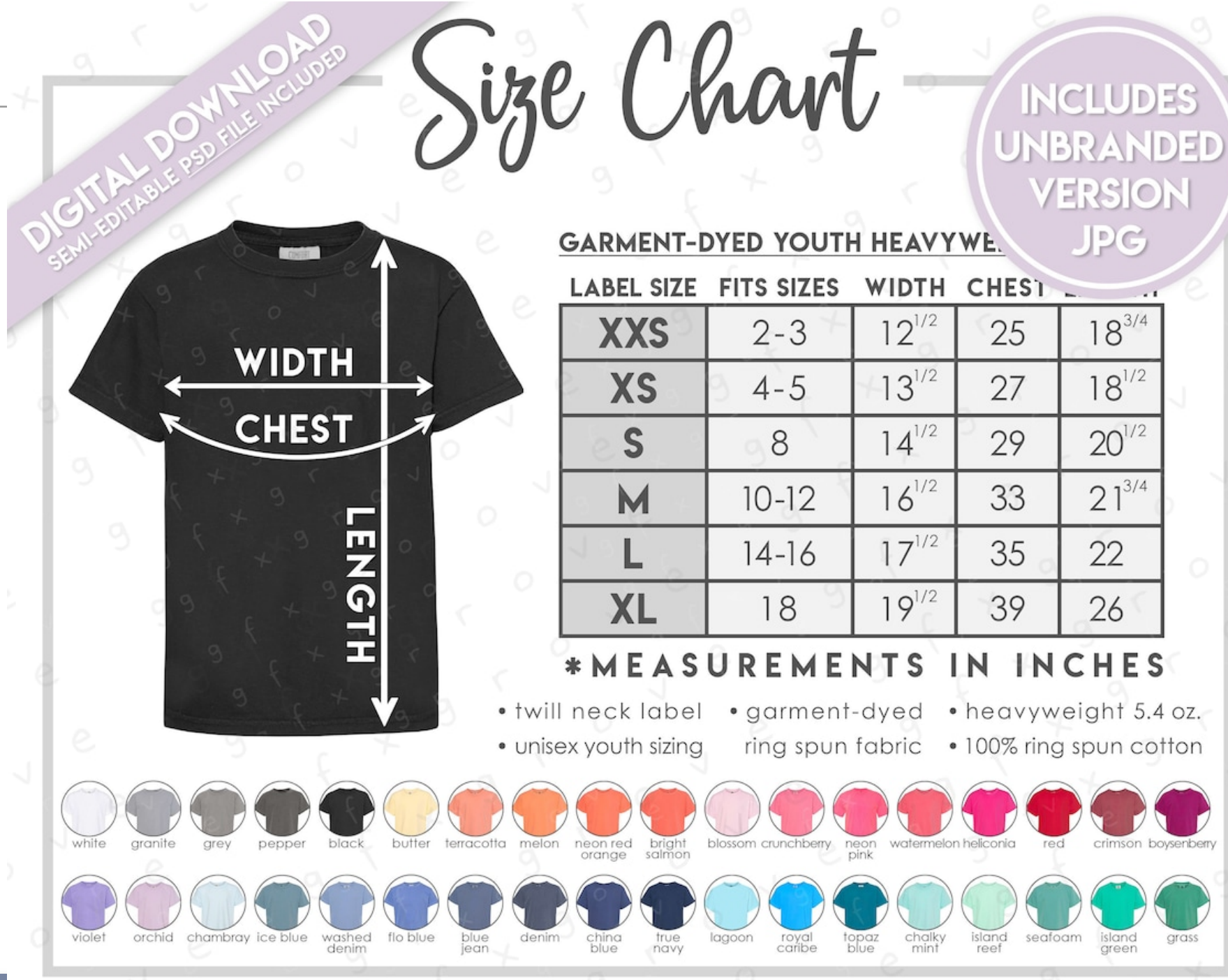


Size (US)	A Length	B Chest
S	27.6"	20.5"
M	28.3"	21.7"
L	29.1"	22.8"
XL	29.9"	24.4"
XXL	30.7"	26.0"
XXXL	31.5"	27.6"
XXXXL	32.3"	29.1"

THESE MEASUREMENTS ARE THE GARMENT MEASUREMENTS.

https://marienicoleclothing.com/pages/mncs-size-chart

Cardinality and Range

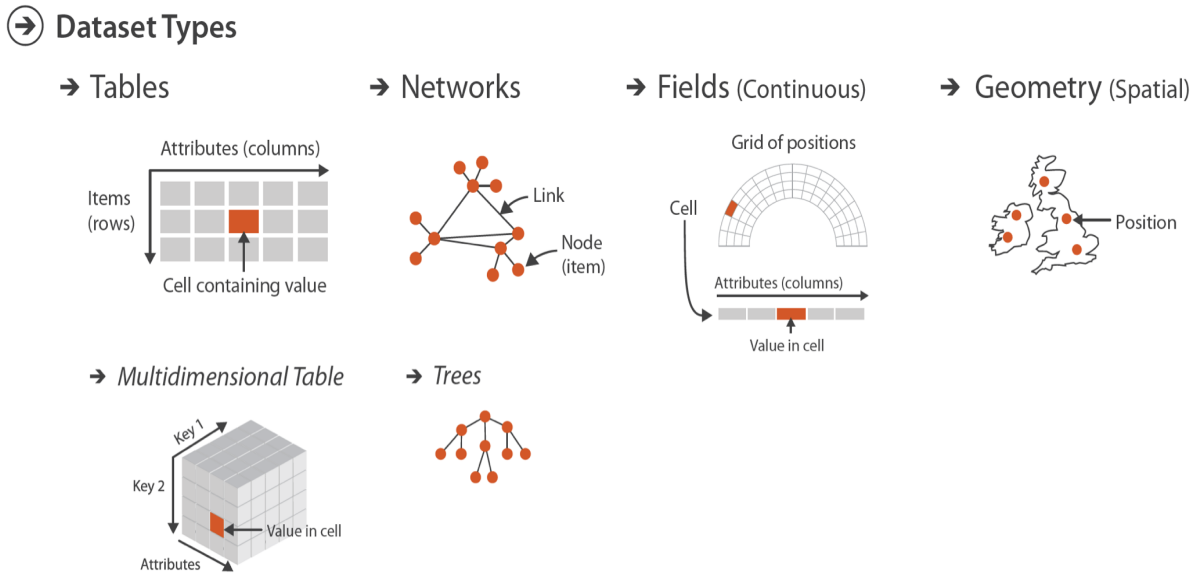


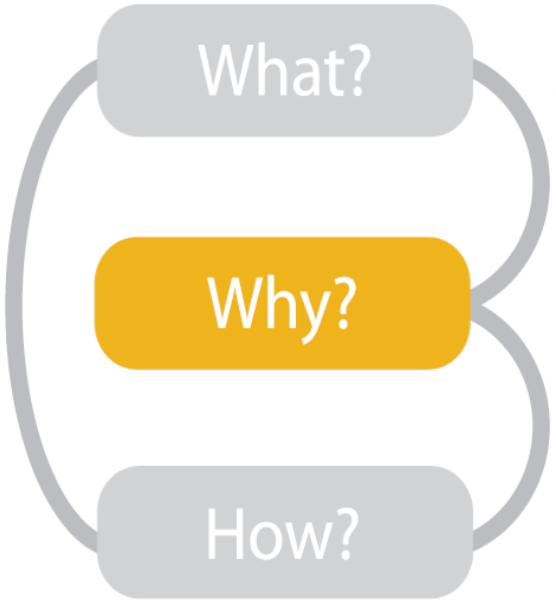
https://i.etsystatic.com/20956615/r/il/6e26c3/4964292245/il_1588xN.4964292245_1xfb.jpg

Quiz

- What are the four main dataset types?

- Networks & Trees
- Wind Tunnel
- Dynamic
- Geometry
- Fields
- Tables
- Websites
- Geology





- {action, target} pairs
  - *Discover distribution*
  - *Compare trends*
  - *Locate outliers*
  - *Browse topology*

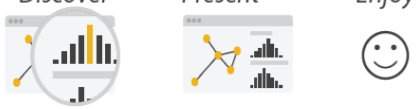
Why?

Actions

Analyze

→ Consume

→ *Discover* → *Present* → *Enjoy*



→ Produce

→ *Annotate* → *Record* → *Derive*

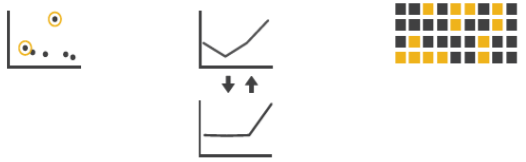


Search

	Target known	Target unknown
Location known	*Lookup*	*Browse*
Location unknown	*Locate*	*Explore*

Query

→ Identify → Compare → Summarize



Targets

All Data

→ Trends → Outliers → Features



Attributes

→ One → Many

→ *Distribution* → *Dependency* → *Correlation* → *Similarity*

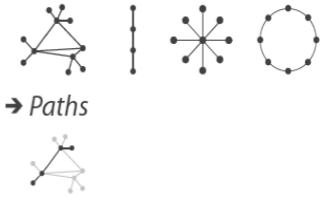
→ *Extremes*



Network Data

→ Topology

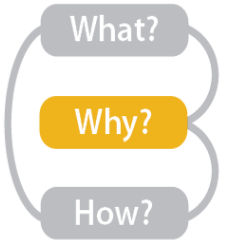
→ *Paths*



Spatial Data

→ Shape





Actions: Analyze

- Analyze → Scrutinize data item(s)
- Search → Find a known/unknown item(s)
- Query → Find out about characteristics of item(s)

- Targets → What is being acted on

High-level Actions: Analyze

Analyze

Consume

→ *Discover*



→ *Present*



→ *Enjoy*



Produce

→ *Annotate*



→ *Record*



→ *Derive*

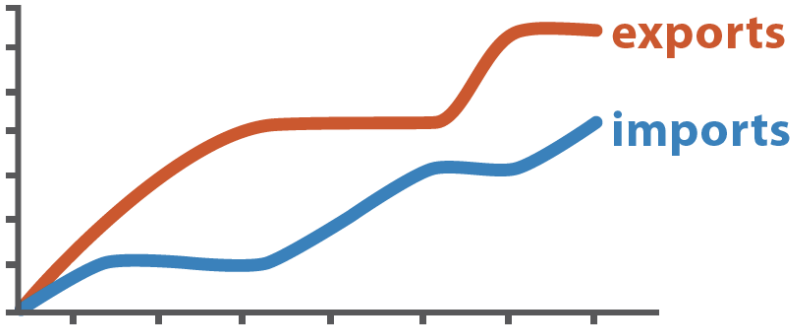


- Consume
  - Discover vs present
    - Classic split aka explore vs explain
  - Enjoy
    - Casual, social

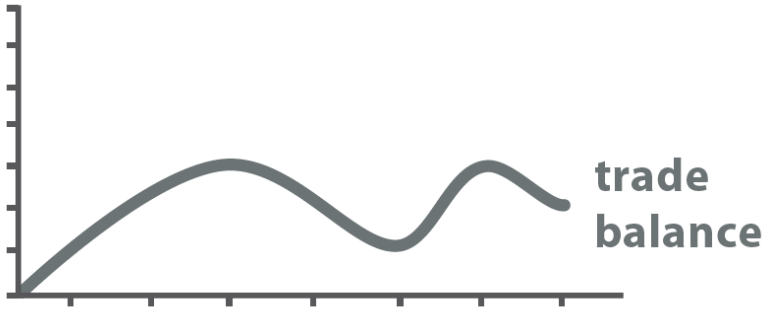
- Produce
  - Annotate, record
  - Derive
    - Crucial design choice

Action: Derive

- Don’t just draw the raw data
- Provide the user with possibilities to transform the data
  - Abstract, reduce, …
- One major strategy for handling complexity



Original Data



trade balance = *exports* – *imports*

Derived Data

Actions: Mid-level Search, Low-level Query

- Search depends on what does user know?
  - Target? Location?

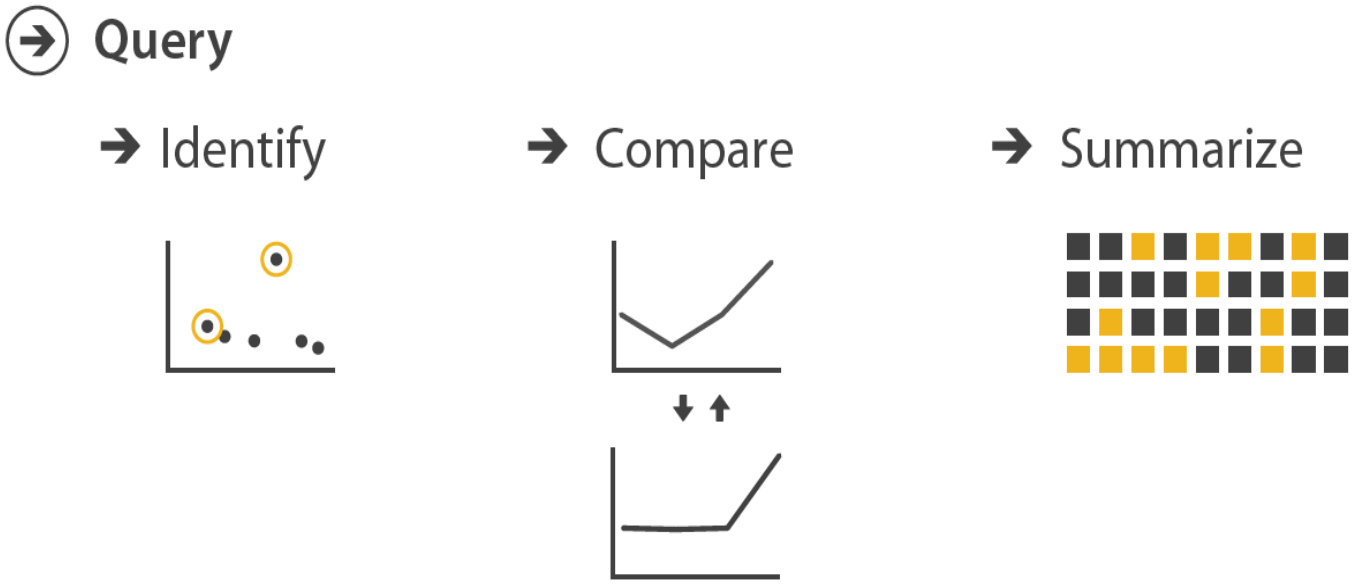
➡ Whiteboard

Search

	Target known	Target unknown
Location known	*Lookup*	*Browse*
Location unknown	*Locate*	*Explore*

Actions: Mid-level Search, Low-level Query

- Query follows search and further refines targets found
  - Identify a single target
  - Compare refers to multiple targets
  - Summarize refers to the full set of possible targets



Why: Targets of Actions

➔ ALL DATA Actions: discover, present, annotate, derive, search, …

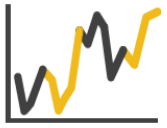
➔ Trends



➔ Outliers



➔ Features



Why: Targets of Actions

ALL DATA

Actions: discover, present, annotate, derive, search, …

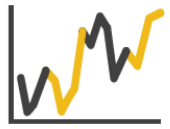
- Trends



- Outliers



- Features



ATTRIBUTES

- One
  - *Distribution*

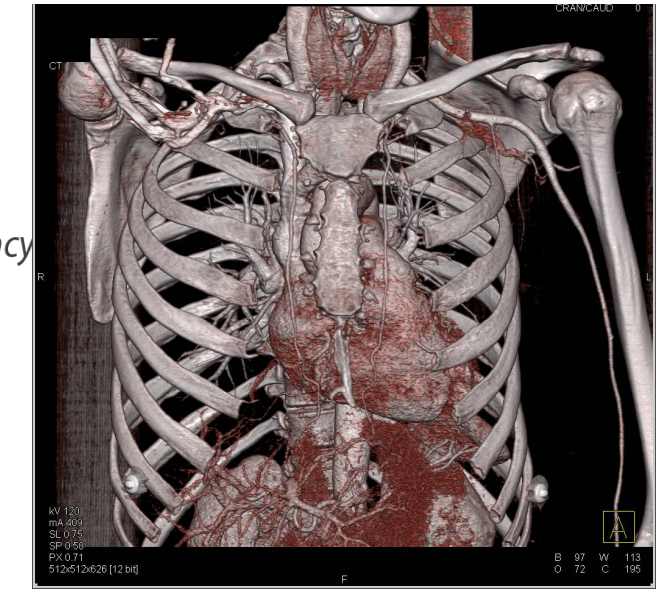


  - *Extremes*



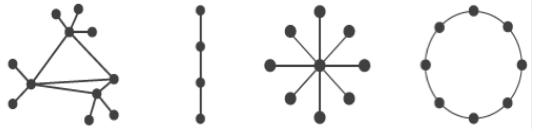
- Many
  - *Dependency*





NETWORK DATA

- Topology



- *Paths*



[SpaceTree: Supporting Exploration in Large Node Link Tree, Design Evolution and Empirical Evaluation. Grosjean, Plaisant, and Bederson. Proc. InfoVis 2002, p 57–64.]

[TreeJuxtaposer: Scalable Tree Comparison Using Focus +Context With Guaranteed Visibility. ACM Trans. on Graphics (Proc. SIGGRAPH) 22:453– 462, 2003.]



Task:
find least common ancestor between nodes/animals in the tree of evolutionary relationships between animals

What? Why? How?

Doing the Abstraction

Domain Task 1:

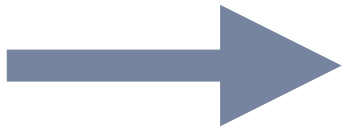
Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Duis faucibus sapien id eros dapibus interdum. Nam mollis dictum posuere.

Domain Task 2:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi laoreet egestas nulla vel elementum.

Domain Task 3:

Sed gravida in tortor vitae tempor. Nam sit amet ullamcorper augue. Curabitur leo elit, tristique eu velit id, interdum porttitor tellus. Ut egestas accumsan ex, sed ultricies est ultrices quis. Donec semper et ipsum malesuada molestie. Maecenas nec scelerisque erat. Suspendisse potenti. Suspendisse nec nibh ac tortor commodo imperdiet sit amet quis dolor.



Task group 1

- Discover Features in entire Table 1: Attribute1, Attributes4
- <Action, Target><Data Set(Subset)><Attributes>

……

Quiz

Drag the words to the correct blanks.

discover _____

_____ trends

locate _____

_____ topology

compare | outliers | distribution | browse

Quiz and End

The design of visualization and interaction techniques can be classified into four high-level techniques:

- [ ] encode
- [ ] manipulate
- [ ] reduce
- [ ] facet
- [ ] color
- [ ] aggregate
- [ ] partition

Quiz

Fill in the missing words

Movie genres are an example of _____ data which has no implicit ordering. On the other hand, ordered data has a well-defined ordering. Weight is an example of _____ data since it allows computation (e.g. twice as heavy). The content rating of movies that determines who is allowed to watch a movie (e.g. PG-13 or R in the US or FSK12/16 in Germany) does not allow arithmetic and is therefore _____.

How?

Encode

Arrange

- Express
- Separate
- Order
- Align
- Use



Map

from categorical and ordered attributes

- Color
  - *Hue*
  - *Saturation*
  - *Luminance*
- Size, Angle, Curvature, ...
- Shape
- Motion

Direction, Rate, Frequency, ...



Manipulate

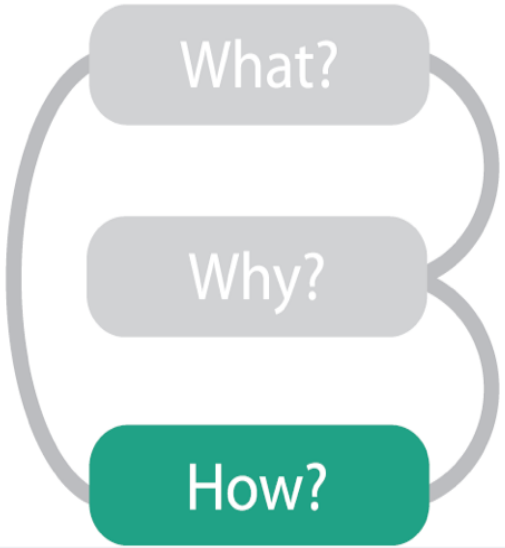
- Change
- Select
- Navigate

Facet

- Juxtapose
- Partition
- Superimpose

Reduce

- Filter
- Aggregate
- Embed



[SpaceTree: Supporting Exploration in Large Node Link Tree, Design Evolution and Empirical Evaluation. Grosjean, Plaisant, and Bederson. Proc. InfoVis 2002, p 57–64.]

[TreeJuxtaposer: Scalable Tree Comparison Using Focus +Context With Guaranteed Visibility. ACM Trans. on Graphics (Proc. SIGGRAPH) 22:453– 462, 2003.]



Task:
find least common ancestor between nodes/animals in the tree of evolutionary relationships between animals

What? Why? How?

Further reading

- Visualization Analysis and Design. Munzner. AK Peters / CRC Press, Oct 2014.
  - Chap 2: What: Data Abstraction
  - Chap 3: Why: Task Abstraction
- A Multi-Level Typology of Abstract Visualization Tasks. Brehmer and Munzner. IEEE Trans. Visualization and Computer Graphics (Proc. InfoVis) 19:12 (2013), 2376–2385.
- Low-Level Components of Analytic Activity in Information Visualization. Amar, Eagan, and Stasko. Proc. IEEE InfoVis 2005, p 111–117.
- A taxonomy of tools that support the fluent and flexible use of visualizations. Heer and Shneiderman. Communications of the ACM 55:4 (2012), 45–54.
- Rethinking Visualization: A High-Level Taxonomy. Tory and Möller. Proc. IEEE InfoVis 2004, p 151–158.

Quiz and End

The design of visualization and interaction techniques can be classified into four high-level techniques:

- [ ] encode
- [ ] manipulate
- [ ] reduce
- [ ] facet
- [ ] color
- [ ] aggregate
- [ ] partition